University of Moratuwa
Faculty of Engineering
Department of Electronic & Telecommunication Engineering

EN 1190: Engineering Design Project
Team LUMOS
Automatic Window Blinds

Group Members

Index No	Name
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Content

1. Problem Description

1.1. Problem

Manual window blinds are still widely used in Sri Lanka, but they present several practical challenges. Elderly, disabled, and physically unwell individuals often struggle to adjust blinds on their own. In large-scale settings such as hotels, offices, and institutions with numerous windows, manual adjustment becomes a time-consuming and labor-intensive task. Additionally, in environments like greenhouses, laboratories, museums and material storage facilities, where maintaining ideal lighting conditions is crucial, manual blinds fail to provide consistent control.

Although motorized and automatic window blind systems with light sensors exist internationally, they are not manufactured locally in Sri Lanka. The limited options available in the local market are imported products that are often prohibitively expensive and not suited for widespread or affordable adoption. As a result, there is a clear gap in the availability of cost-effective, smart window blind solutions that fulfil local needs.

1.2. Expected Goal

Our goal is to provide a practical and accessible solution for individuals who face difficulties in manually operating window blinds, such as the elderly, disabled, or physically unwell, as well as for environments where efficient light control is essential. We aim to simplify and automate the process of adjusting blinds using light-sensing technology, allowing users to maintain optimal indoor lighting conditions effortlessly, without the need for manual labor, technical expertise, or expensive infrastructure

1.3. Solution

Our solution is an Automatic Light-Sensitive Window Blind System designed to adjust the tilt of blinds based on the ambient light level in the room. During installation, users can specify their preferred lighting levels for each room, accommodating different needs such as workspaces, bedrooms, or specialized environments like greenhouses. The system then operates autonomously, maintaining optimal lighting without the need for manual adjustment. Unlike traditional systems, our product is designed to be affordable and user-friendly. We also offer comprehensive after-sales support, including assistance with installation, routine maintenance, and technical guidance to ensure smooth operation over time, which is impossible to get when shipping from overseas.

1.4. Justification of Selection

The lack of accessible and reliable automated window blind systems presents a practical issue in Sri Lanka, especially for individuals with physical limitations and in environments where precise lighting control is essential. Currently, the market relies heavily on imported motorized blind systems, which are not only expensive but also prone to technical failures. Moreover, these systems are often limited to premium establishments and are largely unavailable to the general public.

Through discussions with local window blind sellers and industry professionals, we identified a significant gap in the Sri Lankan market for an affordable, durable, and locally supported automated window blind solution. Sellers expressed a willingness to collaborate, noting that a reliable, cost-effective alternative would benefit both residential and commercial customers. Our proposed system directly addresses these concerns by offering a customizable, light-sensitive solution backed by local after-sales support.

2. Product Feasibility

2.1. Technical Feasibility

We have successfully developed a technically viable prototype using reliable and accessible components. To measure indoor light intensity, we used two BH1750 light sensors to obtain an average reading, which helps minimize the impact of sudden light spikes and ensures accurate decision-making. The system is controlled by an ESP-WROOM-32E microcontroller, which processes sensor inputs and adjusts the blinds accordingly.

For actuation, we utilized the 28BYJ-48 stepper motor, which precisely controls the angle of the window blinds. The system is powered through an AC supply; regulated via a 12V power pack to ensure consistent and stable performance and a voltage regulator to maintain a stable voltage at 3.3V. All necessary components were acquired, and the integration was completed successfully. The enclosure and the connecting shaft were 3D printed using PLA. We designed and assembled the circuit ourselves, including custom soldering work and mechanical alignment for the blind mechanism.

2.2. Hardware Feasibility

The hardware components we used are

- ESP-WROOM-32E Chip
- BH1750 Light Intensity Sensors
- Voltage Regulator Module
- Stepper Motor 28BYJ-48
- LED Indicators
- 12V Power Pack

2.3. Software Feasibility

Software	Programming Language	Purpose
Arduino	C++	To Program the ESP 32 chip
SOLIDWORKS	–	To Design the Enclosure
Firebase Studio	React	To Create the Web App

3. Product Applications

The main application of the automatic light-sensitive window blind system is to maintain optimal indoor lighting conditions without manual intervention. By continuously monitoring the ambient light levels using dual BH1750 sensors, the system automatically adjusts the tilt angle of the blinds to match the user-defined brightness preferences set during installation. This ensures a consistent and comfortable lighting environment across different rooms, while also improving energy efficiency and reducing the need for frequent manual adjustments, especially beneficial for elderly, disabled individuals, and large-scale settings such as hotels and offices.

LUMOS Blinder Web Application

The LUMOS Smart Blind Web application is connected to the device via Wi-Fi. It displays real-time ambient light levels and blind angles through an interactive dashboard. Users can switch between automatic and manual control modes, view historical data through dynamic charts, and receive AI-generated insights about lighting patterns.

The app also allows configuration of system settings like sensing intervals and includes a responsive UI with mock authentication for user-specific access.

Core Technologies

- Frontend Framework: Next.js 15 with the App Router, enabling a modern, server-first web application with optimized performance.

- UI Library: React 18 for building interactive and stateful user interface components.

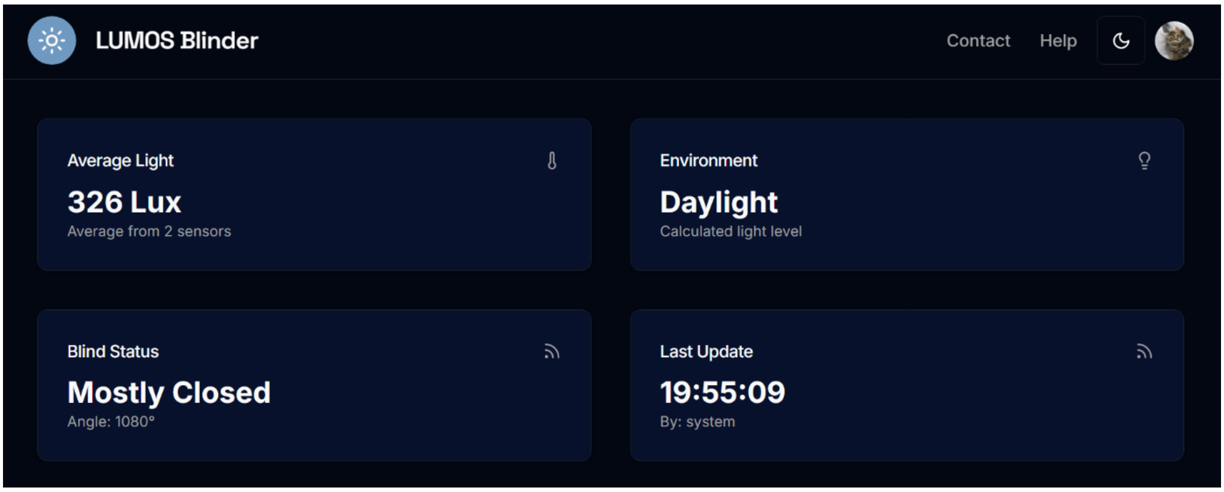
- Styling: Tailwind CSS for a utility-first styling approach, combined with ShadCN UI for a pre-built, accessible, and themeable component library.

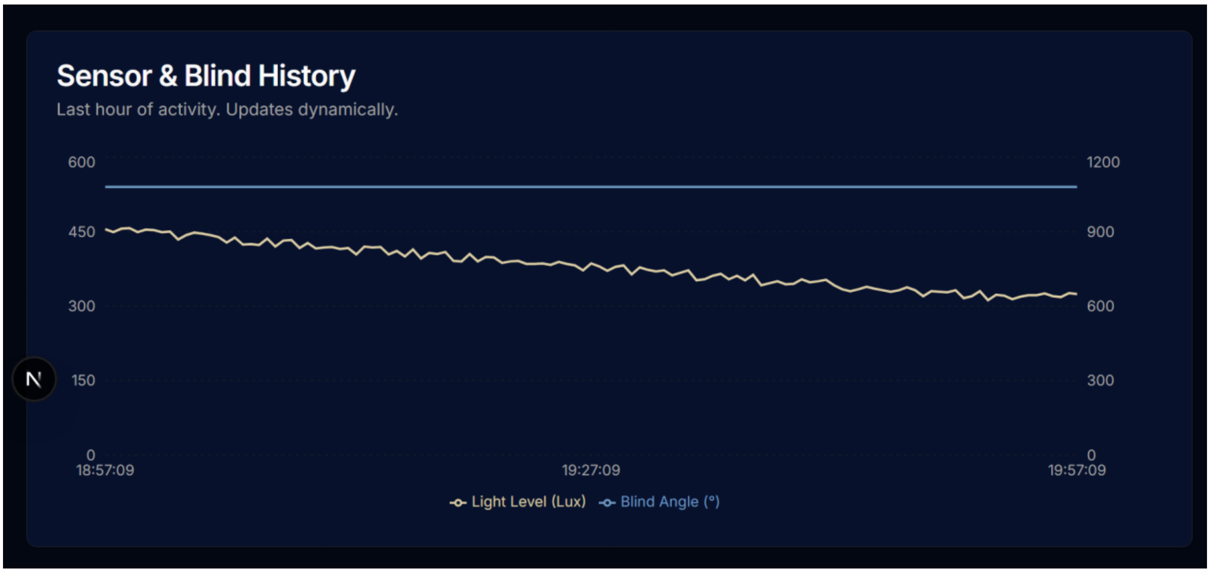
- Artificial Intelligence: Google's Genkit framework to integrate generative Al, powering the "Al Assistant" feature for data analysis and insights.

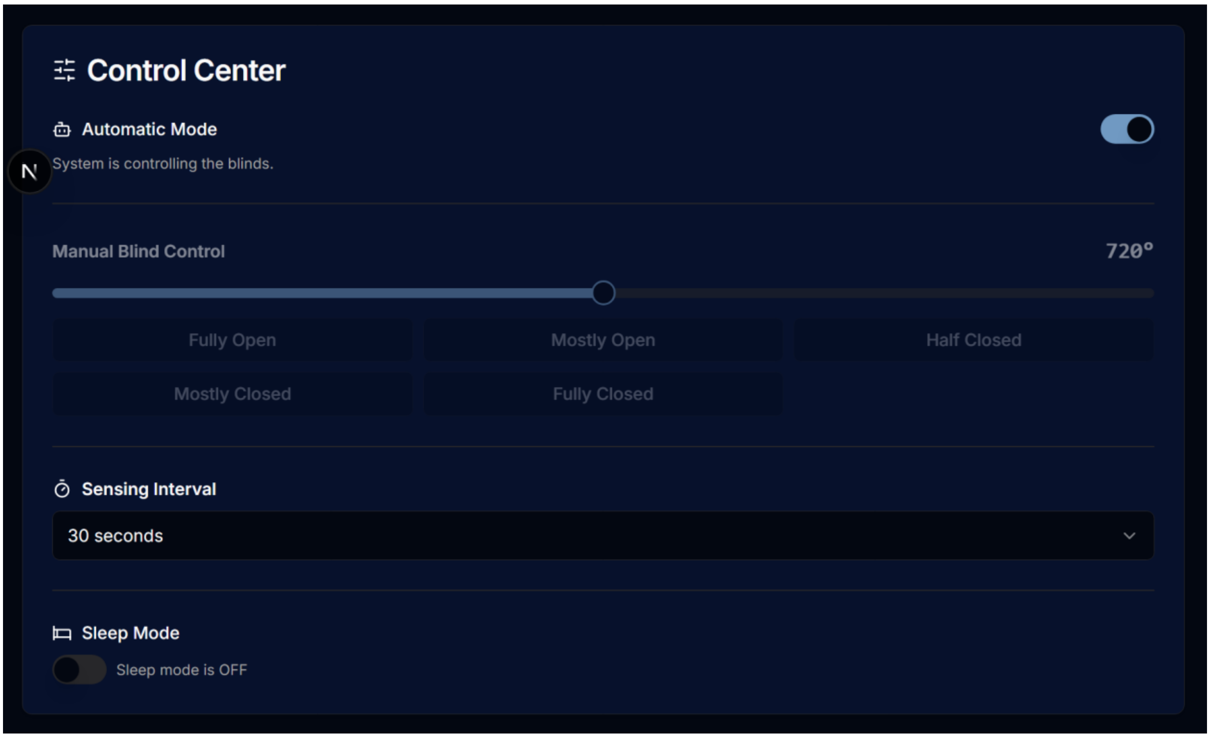
- Language: TypeScript for type safety, improved developer experience, and more maintainable code.

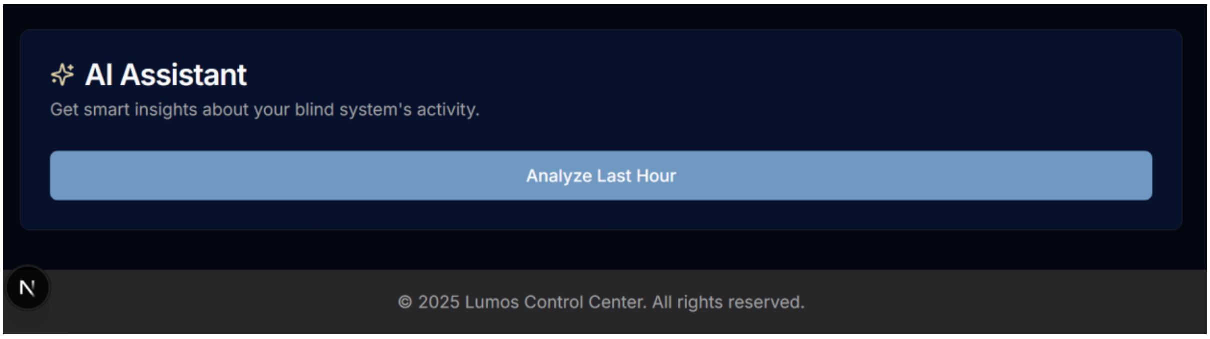
Key Functionalities

- Real-time loT Dashboard: The application features a live dashboard that simulates real-time data from lot sensors (lux sensors) and actuators (window blinds).
- Dynamic Data Visualization: A responsive line chart visualizes historical sensor and blind angle data, updating dynamically to reflect system changes.
- Dual Control Modes:
    1. Automatic Mode: The system intelligently adjusts the blind angle based on simulated ambient light levels, mimicking a real- world automated smart home device.
    2. Manual Mode: Users can disable the automatic system and take direct control of the blinds using a slider and preset position buttons.
- Al-Powered Insights: An "Al Assistant" uses a Genkit flow to analyze the recent history of sensor data and blind movements, providing users with easy-to-understand summaries and insights about the system's behavior.
- System Configuration: Users can configure system parameters, such as the Sensing Interval, to simulate changing the device's data reporting frequency. A "Sleep Mode" toggle is also included as a UI element.
- Responsive and Modern UI: The user interface is designed to be fully responsive, adapting seamlessly to desktop, tablet, and mobile screen sizes. It also includes a dark/light mode theme toggle for user preference.
- Mock Authentication: The app includes a simulated user authentication flow to demonstrate how UI elements (like controls) can be enabled or disabled based on user sign-in status.




Sensor & Blind History
Last hour of activity. Updates dynamically.
600
450
300
150
0
1200
900
600
300
0
18:57:09
19:27:09
19:57:09
Light Level (Lux)
Blind Angle (°)


Control Center
Automatic Mode
System is controlling the blinds.
Manual Blind Control
720°
Fully Open
Mostly Open
Half Closed
Mostly Closed
Fully Closed
Sensing Interval
30 seconds
Sleep Mode
Sleep mode is OFF


AI Assistant
Get smart insights about your blind system's activity.
Analyze Last Hour
© 2025 Lumos Control Center. All rights reserved.

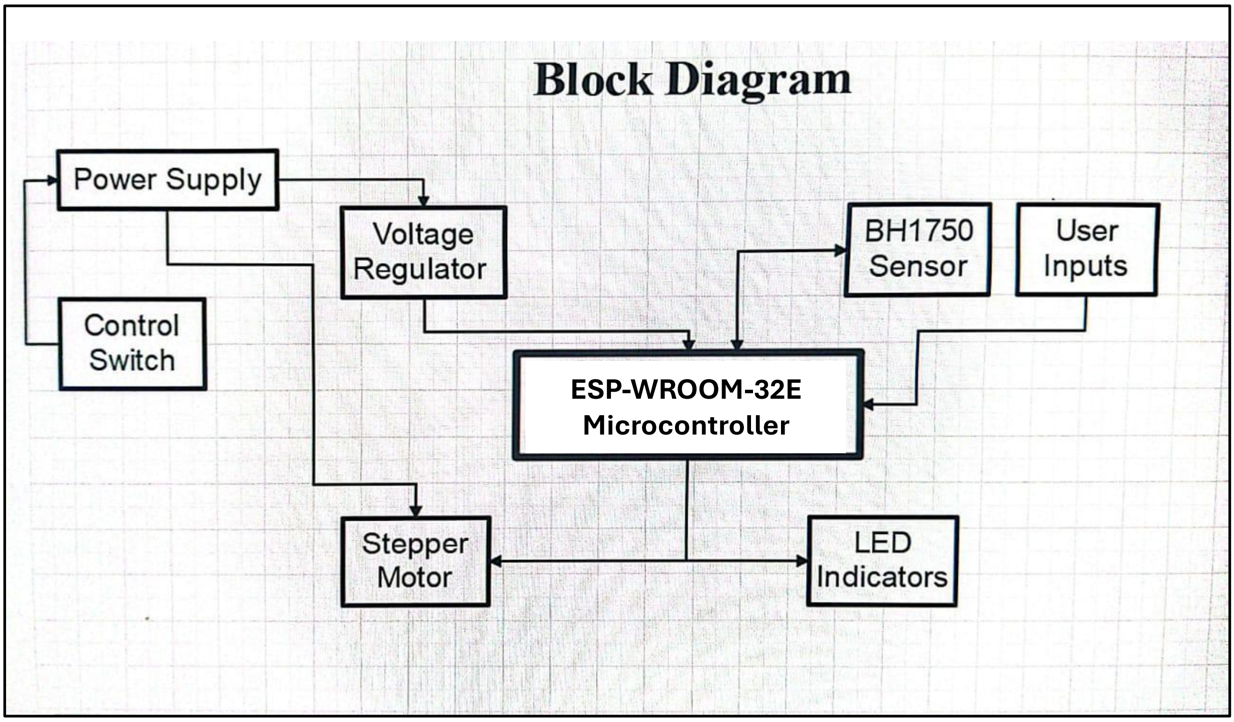
4. Product Specifications

- Type: Automatic Window Blind Controller with Light Sensors
- Synonyms: Smart Blind Controller, Light-Sensitive Blinds, Motorized Smart Blind Adjustment System
- Control Method: Light Intensity-Based Automatic Adjustment
- Power Supply: 12V DC via AC Power Pack
- Dimensions: 140x50 mm
- Weight: 150g
- Connectivity: Wired
- Installation: Wall-Mountable Control Box
- User Settings: Customizable Brightness Thresholds per Room at Installation

5. Product Architecture

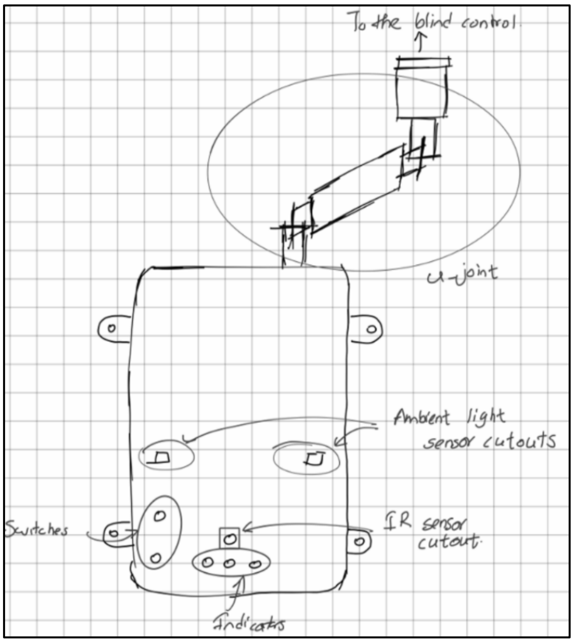
- Power Supply - The system is powered through a 12V DC supply using an external AC power pack.
- ESP 32 chip - The ESP-WROOM-32E microcontroller is used to process real time light intensity data from the sensors and control the stepper motor accordingly. It handles averaging, decision making logic, and motor control.
- Voltage Regulator - Regulates the input voltage to 3.3V for the ESP32 to operate reliably.
- Light Sensors - Two BH1750 digital light sensors are used to measure indoor light intensity. Averaging the readings from both sensors helps avoid errors caused by sudden light fluctuations and improves the accuracy of blind adjustment.
- Motor - The 28BYJ-48 stepper motor is used to adjust the tilt angle of the window blinds smoothly and precisely.
- Web App – Connected to the device via Wi-Fi, displays light levels and blind angles, and allows the users to switch between manual and automatic control modes.

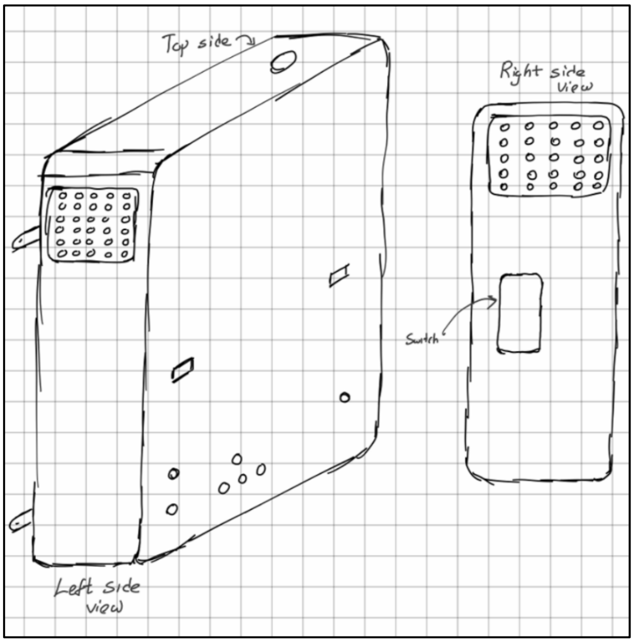
5.1. Block Diagram



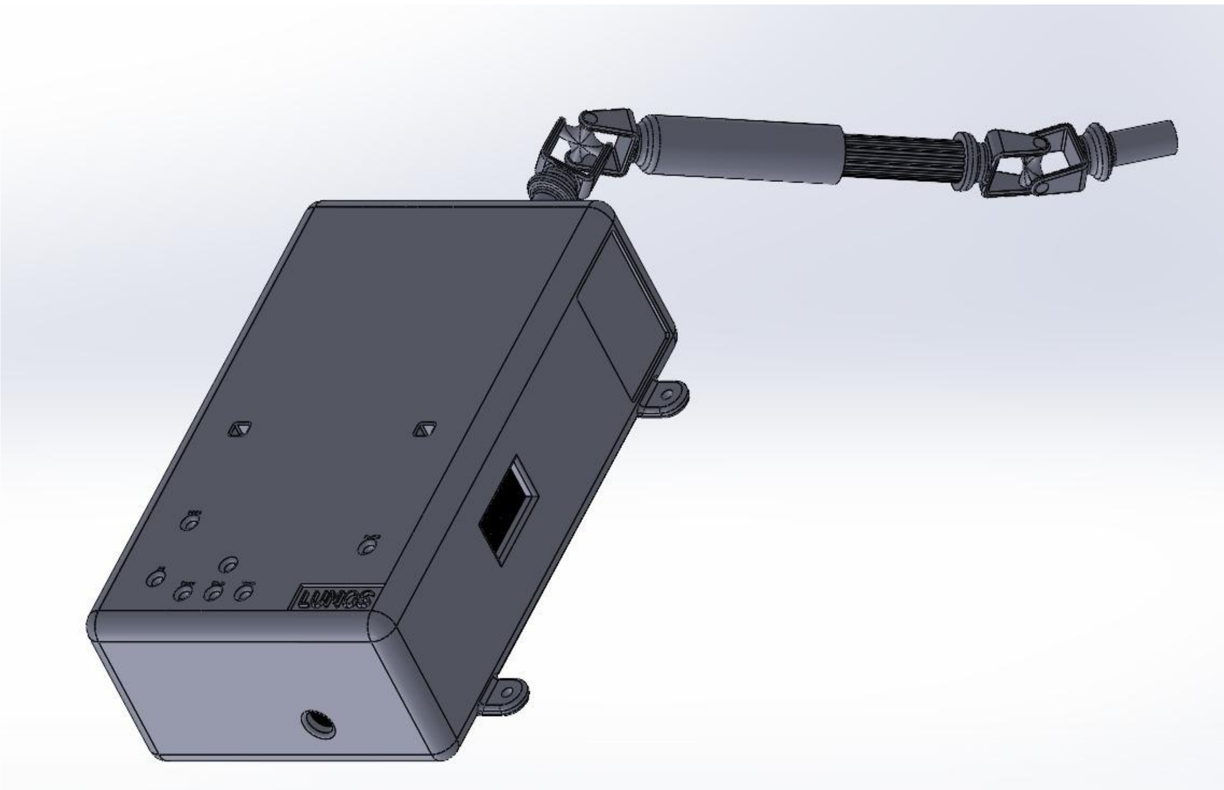
6. Enclosure Design

6.1. Initial Sketches

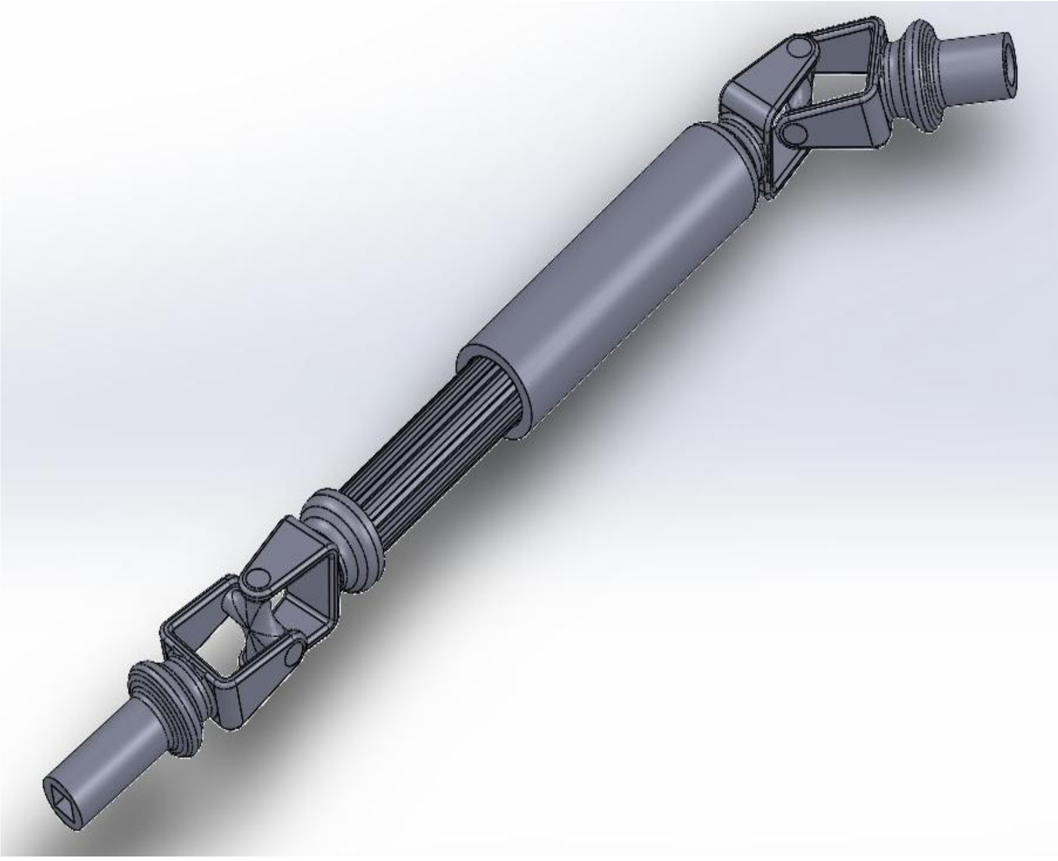




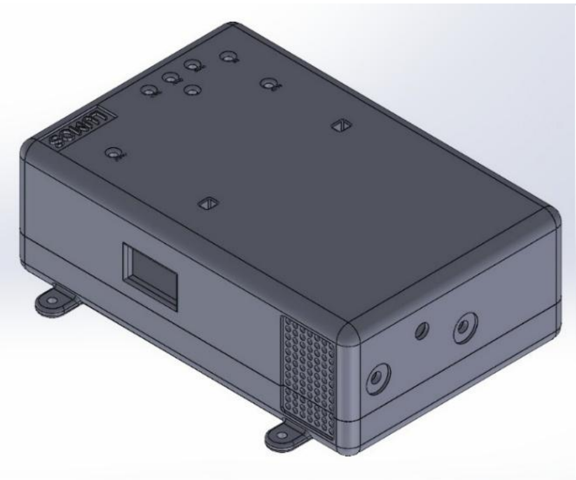
6.2. Final Sketch

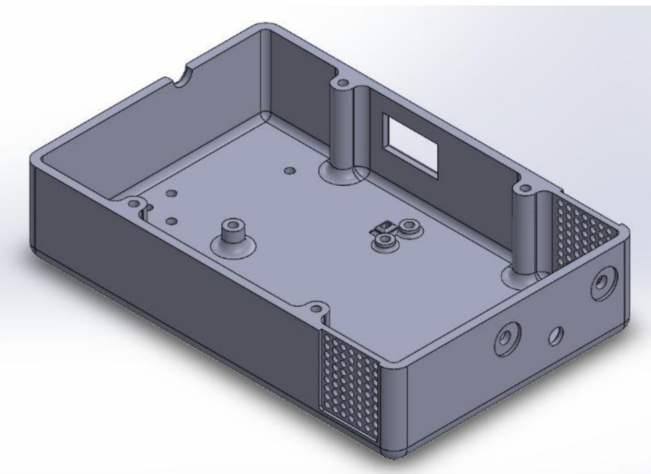


6.3. Connecting Mechanism (Shaft)

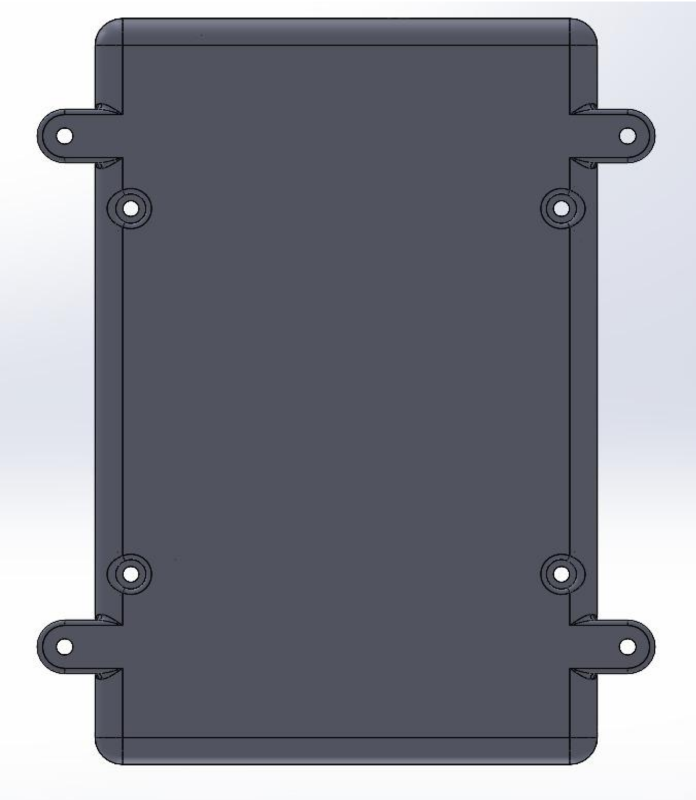


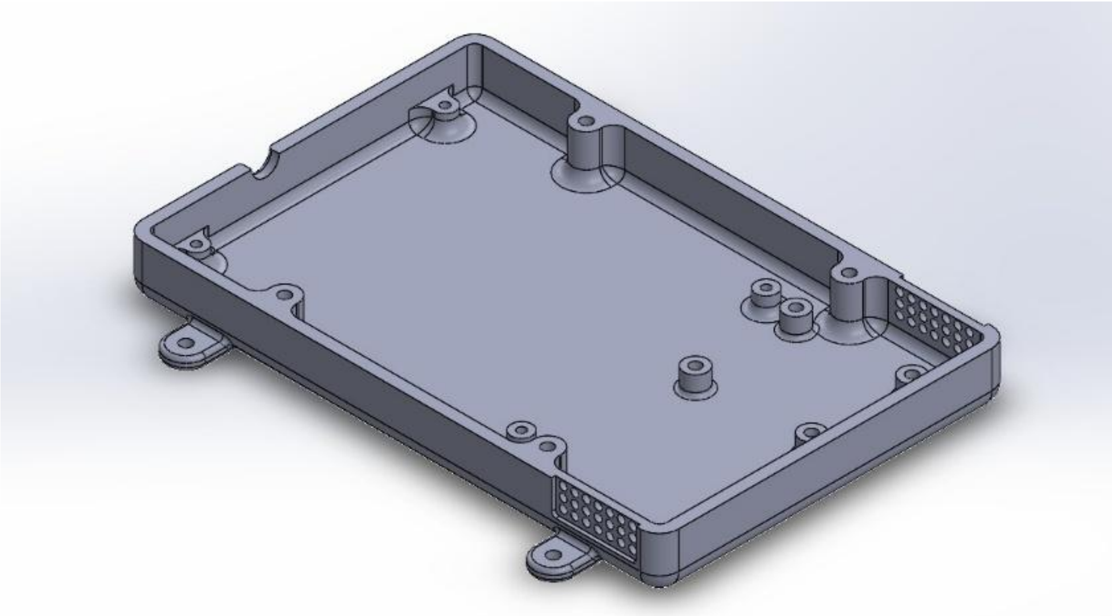
6.4. Top View



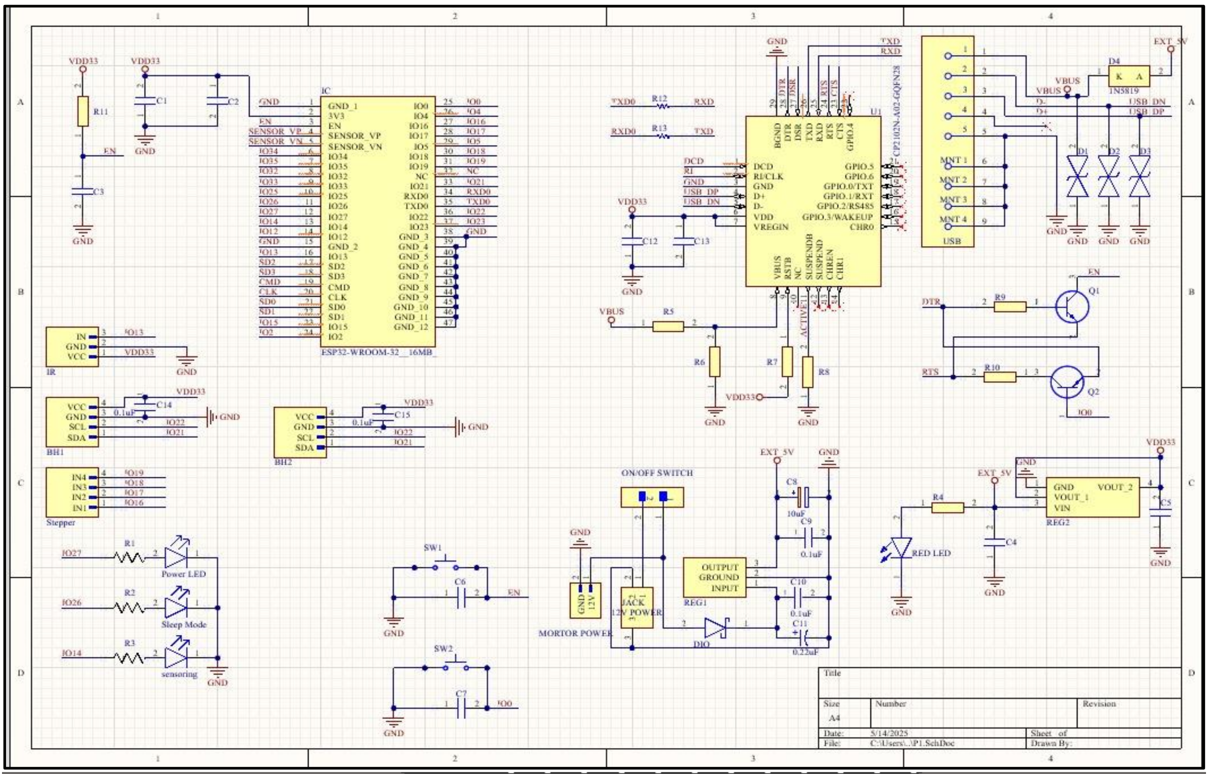


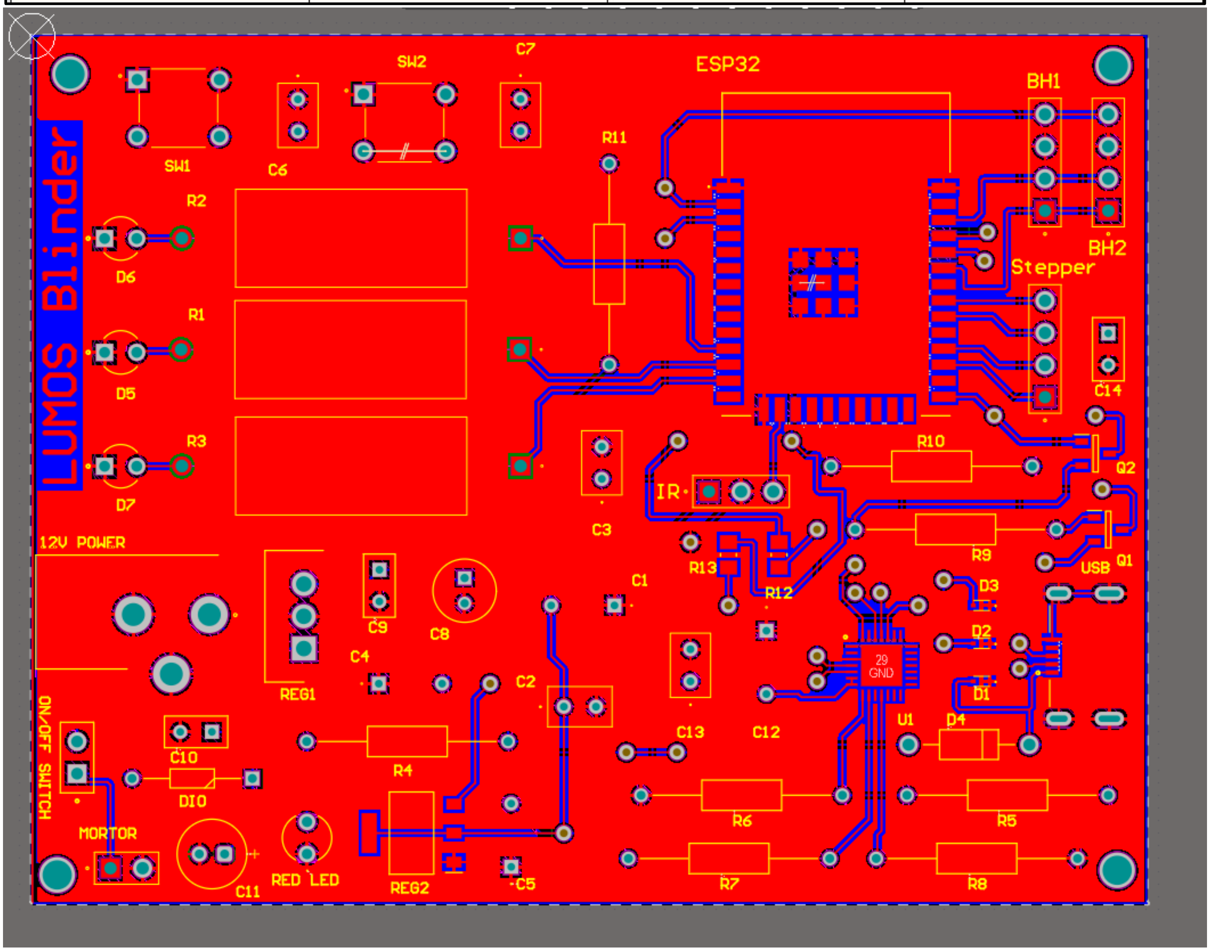
6.5. Bottom View

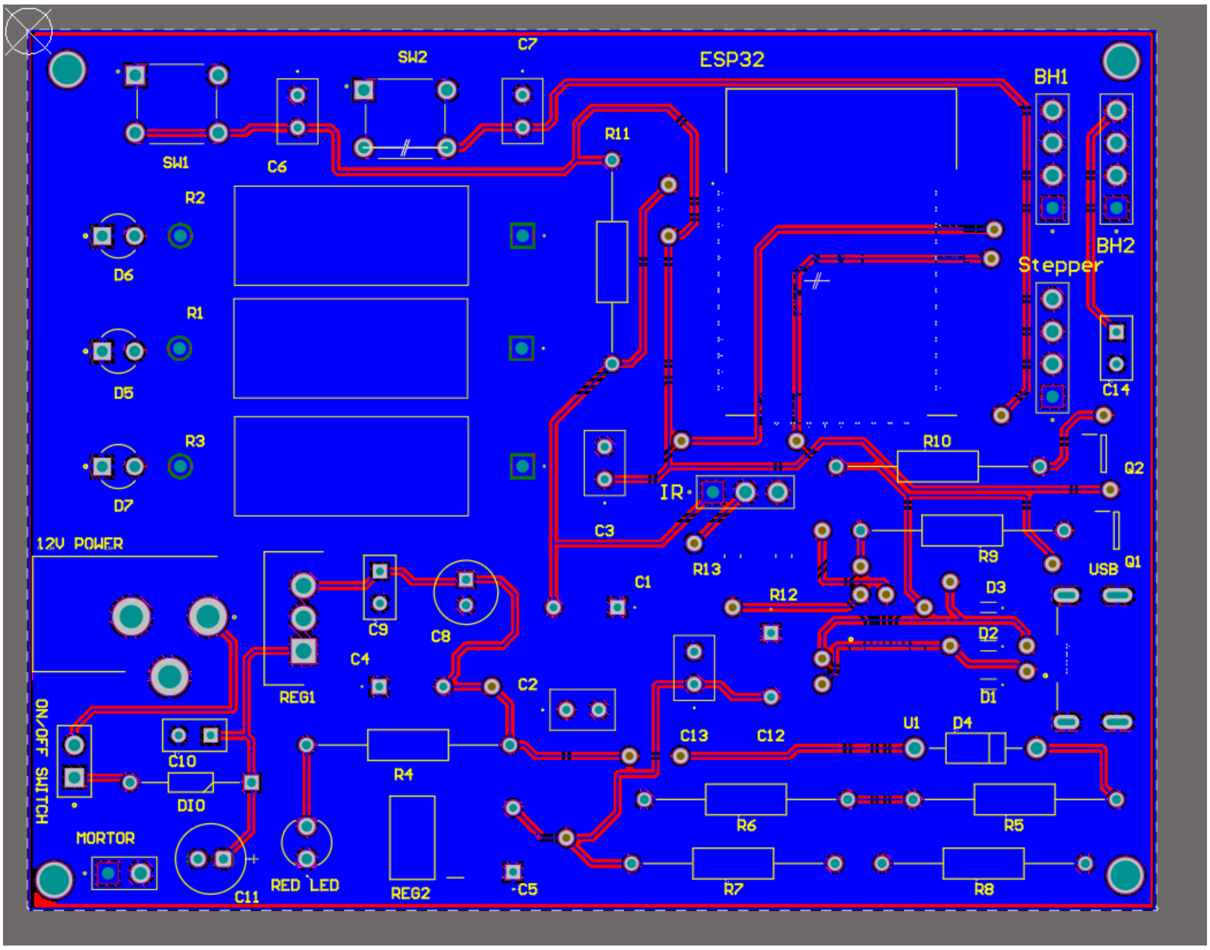




7. PCB Design




LUMOS Blinder
ESP32
SW1
SW2
C6
C7
R11
BH1
BH2
Stepper
R2
D6
R1
D5
R3
D7
C14
R10
Q2
IR
C3
R9
Q1
USB
R13
R12
D3
D2
D1
C1
12V POWER
REG1
C9
C8
C4
C2
U1
D4
C13
C12
ON/OFF SWITCH
C10
DIO
R4
MORTOR
C11
RED LED
REG2
C5
R6
R5
R7
R8


ESP32
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C14
R10
Q2
IR
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R13
R9
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Q1
C1
R12
D3
D2
D1
12V POWER
C9
C8
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C2
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C10
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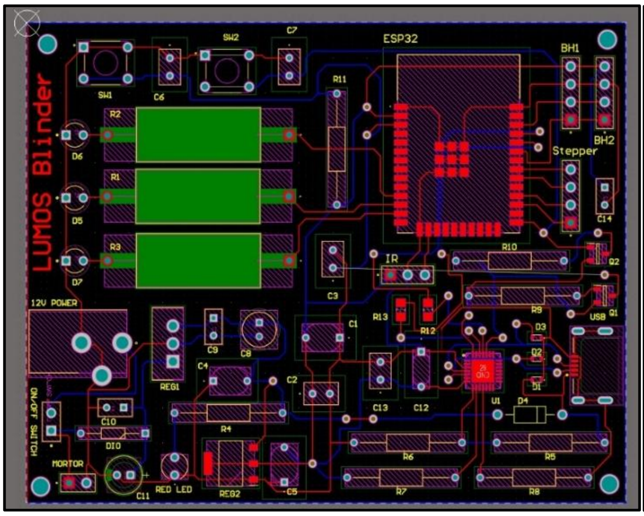
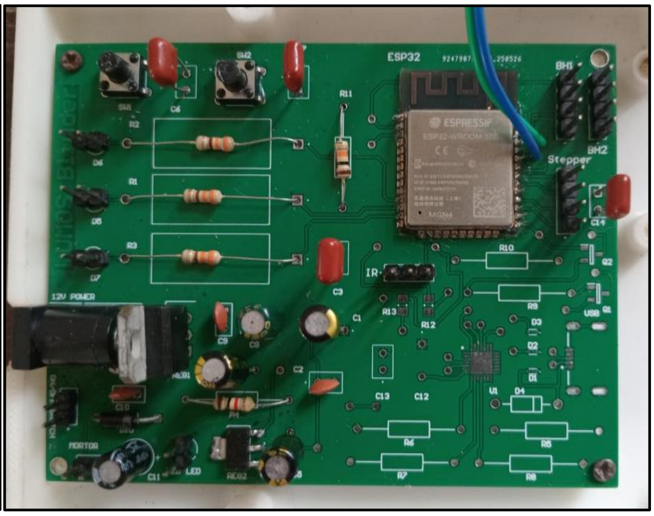
C11
RED LED
REG2
C5
R7
R8


LUMOS Blinder
ESP32
BH1
BH2
Stepper
12V POWER
ON/OFF SWITCH
MORTOR
RED LED
REG2


ESP32
BH1
BH2
Stepper
12V POWER

8. Final Product

9. Market Analysis

The smart blind automation market is gaining momentum globally due to increasing demand for energy efficiency, convenience, and smart home integration. Automated blind systems equipped with light sensors offer benefits such as optimizing natural lighting, reducing air conditioning use, and improving indoor comfort, all without user intervention. These systems are particularly useful in homes, offices, hospitals, and hotels, and also support accessibility for elderly or disabled individuals who may find manual blind operation difficult.

Our target audience includes environmentally conscious homeowners, tech-savvy consumers, corporate offices, hospitality businesses, and interior designers who seek modern, minimal-effort solutions. Possible business models for our product include direct-to-consumer sales, partnerships with hardware retailers and interior design firms, and even customized B2B deals for bulk deployment in institutions like hotels or hospitals.

However, we identified that there are no local manufacturers offering such systems in Sri Lanka, despite the rising interest in home automation. As a result, consumers must rely on international sellers, with base prices starting around LKR 21,000, but reaching up to LKR 33,000 to 36,000 after adding shipping and customs fees. Additionally, most of these imported products lack local warranty, maintenance, or after-sales service—resulting in inconvenience and higher lifetime costs.

This clear market gap highlights a strong opportunity for a locally developed, cost-effective blind automation solution with light sensoring that can be easily installed without replacing existing blinds. Challenges such as educating the public, building brand credibility, and competing with well-established global brands remain, but overcoming these barriers will be essential for capturing a significant share of this emerging market in Sri Lanka.

10. Cost Analysis and Pricing

10.1. Project Budget with BOQ

EXPENSE	Unit Price (Rs.)	Quantity	COST (Rs.)
Enclosure	4,850.00	1	4,850.00
PCB	3,360.00	1	3,360.00
ESP-WROOM-32E Chip	1,000.00	1	1,000.00
Stepper Motor 28BYJ-48	600.00	1	600.00
BH1750 Sensors	125.00	2	250.00
RLC components & Wiring			400.00
Connector Shaft Mechanism	750.00	1	750.00
Voltage Regulator + USB module			500.00
12V Power Pack	500.00	1	500.00
Misc.			200.00
TOTAL			**12,410.00**

10.2. Pricing Justification

The selling price for the product was determined by first calculating the manufacturing cost, which totals approximately Rs. 12,410 per unit. This includes all electronic components, motor, power supply, enclosure, and assembly. A reasonable markup was then applied to ensure profitability, after-sales service capability, and to account for future reductions in cost through mass production.

Considering these factors, the estimated selling price was set at **Rs. 18,000 to Rs. 20,000**, which remains significantly lower than the Rs. 33,000–36,000 required to import similar products. This pricing provides customers with savings of up to Rs. 20,000, while also offering local support and maintenance—something that imported alternatives lack. This ensures both affordability and long-term usability for Sri Lankan consumers.

11. Future Improvements

Introduction to Solar Power for Eco-Friendliness

Implementing solar panels to power the system, reducing dependence on grid electricity and promoting sustainability.

Integration of IoT Features

Adding Wi-Fi or cloud connectivity for remote control, real-time monitoring, and integration with smart home systems.

Multiple Blind Synchronization for Efficiency

Enabling control of multiple blinds simultaneously to maintain uniform lighting across large rooms or buildings.

Design Configurations for Other Types of Blinds

Adapting the system to support vertical blinds, roller blinds, and curtain mechanisms for broader compatibility

12. Contribution Report

Member	Tasks
Eranga W.A.O. (230175U)	Circuit & PCB Design Microcontroller Programming & Testing Web App Design
Gamage S.K. (230195F)	Product Idea, Hardware Selection Survey Analysis & Presentation
Samarasinghe S.M.R.R. (230566U)	Enclosure Design & Optimization Product Testing & Debugging
Tharushika G.K.E. (230636K)	Product Assembly & Testing Market Analysis and Budget Calculation